Mansi Bhardwaj

Embedded Systems Engineer - Robotics & Instrumentation
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Profile

MSc Embedded Systems student at the University of Twente with 2+ years of industry experience in embedded hardware, firmware, and real-time sensing systems for ROV/AUV platforms. Experienced in embedded electronics, PCB design, multi-sensor integration, signal integrity, and firmware development on ARM/AVR platforms. Interested in robotics, instrumentation, FPGA systems, and high-reliability engineering systems.

Education

University of Twente **Sept 2025 - Present**
Master of Science in Embedded Systems *Enschede, Netherlands*

Vellore Institute of Technology **July 2019 - Aug 2023**
B.Tech in Electronics and Communication Engineering *Chennai, India*

Experience

Planys Technologies **July 2024 - August 2025**
Design and Integration Engineer, Chennai, India - Full-time

- Designed embedded hardware and firmware for ROV/AUV systems including power management, sensor interfaces, and communication subsystems.
- Developed real-time multi-sensor acquisition systems on ARM/AVR microcontrollers using SPI, I^2C, and UART communication protocols.
- Designed and validated 4-layer PCBs with controlled impedance routing, signal integrity considerations, and EMI-aware layout practices for deployment in harsh marine environments.
- Performed board bring-up, debugging, hardware validation, and signal integrity analysis using laboratory instrumentation including oscilloscopes and LCR meters.

Zaptics Research **June 2023 - July 2024**
Junior Engineer (Embedded Systems), Noida, India - Full-time

- Designed schematics and high-speed embedded hardware in Altium Designer for performance-critical embedded systems.
- Developed optimized 4-layer PCB layouts with controlled impedance routing, robust power distribution, and signal integrity considerations.
- Contributed to board bring-up, hardware debugging, validation testing, and component optimization for embedded systems.

Technical Skills

Embedded Systems: ARM (STM32), AVR (ATmega), Embedded C/C++, RTOS Concepts, SPI, I^2C, UART, ADC/DAC, Interrupt-Driven Systems
Robotics & Instrumentation: Multi-Sensor Integration, Real-Time Data Acquisition, Embedded AI, Signal Processing, Hardware Validation
Hardware & PCB Design: Altium Designer, Eagle, 4-Layer PCB Design, Controlled Impedance Routing, Signal Integrity, EMI/EMC Mitigation, Board Bring-Up
FPGA & Digital Design: Verilog, VHDL, FSM Design, Timing Analysis, SoC Architectures, Vitis-AI
Programming & Tools: C, C++, Python, MATLAB, Git, LTspice, TINA-TI, Proteus, Microchip Studio, Linux
Languages: English (Fluent), Dutch (A1 - Currently Learning)

Projects

Wearable Ergonomic Risk Awareness System **2026**

- Developed a real-time wearable posture monitoring system using ESP32-S3, MPU6050 IMU sensors, embedded haptic feedback, FSM-based posture classification, and INT8 neural network inference.
- Implemented synchronized multi-sensor I^2C architecture using TCA9548A multiplexer for real-time signal acquisition and processing.

FPGA-Accelerated Embedded AI for Resource-Constrained Systems **2026**

- Designed and optimized CNN-based image classification networks for FPGA-accelerated embedded AI deployment on the KRIA KV260 platform using Vitis-AI and INT8 quantization.

Detection of Parkinsonian Syndrome using Wearable Device **2021**

- Designed wearable sensing system integrating EMG and IMU sensors; implemented signal processing and feature extraction pipeline achieving 92% classification accuracy.

Automatic Pet Feeder - IoT Based Embedded System **2021**

- Designed low-power IoT-based embedded control system with wireless communication; published in AIP Conference Proceedings (DOI: 10.1063/5.0151143).